MD ROKNUZZAMAN RIKON

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RESEARCH INTERESTS

Wearable & Metasurface-Integrated Antennas; Reconfigurable and Flexible Antenna Systems; Machine Learning-Assisted Antenna Design and Optimization; Antenna Arrays & Beamforming

EDUCATION

B.Sc. in Avionics Engineering **July 2022 – July 2026**

Aviation and Aerospace University, Bangladesh (AAUB)

- CGPA: 3.37/4.00

PUBLICATION

"Design and Machine Learning-Assisted Optimization of a Compact Frequency-Reconfigurable Trophy-Shaped Microstrip Antenna for Multifunctional Communication," Md. Abu Bakar Siddik, **Md Roknuzzaman Rikon**, Shahnaz Malik, Tamzidul Hoque Tahmid, Sabrina Nowrin Mim. — presented at 2026 IEEE International Conference on Microwave, Antennas, RF and Sensors (MARSCON), AUST, Dhaka, Bangladesh, Aug. 13-14,2026

- Developed and evaluated machine-learning models (Random Forest and Gradient Boosting Regression) for antenna performance prediction and optimization, reducing the need for repeated full-wave electromagnetic simulations.

Manuscript in preparation: "Dual-Band Metasurface-Backed Wearable Logo-Shaped Antenna for Pilot Health Monitoring," **Md Roknuzzaman Rikon**, Md Rashid Karim, Md. Abu Bakar Siddik, Sakib Hossain Porosh — to be submitted to *Progress In Electromagnetics Research (PIER)*

RESEARCH EXPERIENCE

Research Assistant, Department of Avionics Engineering, AAUB **Aug 2026 – Present**

Supervisor: Head, Department of Avionics Engineering

- Working on antenna array design and beamforming using CST Studio Suite and MATLAB, focusing on directional radiation and beam steering.
- Investigating element spacing and phase excitation for radiation-pattern control, beam steering, and sidelobe reduction.
- Exploring simulation-based optimization of antenna-array parameters for improved wireless sensing and communication performance.
- Exploring machine-learning-assisted optimization approaches for adaptive wireless systems and sensing applications.

Undergraduate Thesis Researcher, AAUB **Jul 2025 – Jul 2026**

Supervisor: Md. Abu Bakar Siddik, Assistant Professor, Dept. of Avionics Engineering

- Thesis: Designed and simulated a dual-band (2.45/5.8 GHz) metasurface-integrated wearable microstrip antenna in CST Studio Suite, focusing on antenna optimization, radiation characteristics, and SAR analysis for aviation and healthcare applications.
- Performed parametric studies to evaluate the effects of antenna and metasurface design parameters on resonant frequency, impedance matching, and radiation performance.

Research Focus: AI/ML-assisted antenna optimization, flexible and reconfigurable antenna systems, wireless sensing, beamforming, and next-generation wireless communications.

TECHNICAL SKILLS

- **Programming & Data Analysis:** Python, MATLAB, C/C++
- **Machine Learning:** Random Forest, Gradient Boosting Regression, Regression Modeling, ML-assisted Optimization
- **Signal Processing:** Signal Acquisition, Filtering, Signal Analysis, Digital Signal Processing
- **Wearable & RF Systems:** Wearable Antennas, Microstrip Antennas, Metasurface/AMC, SAR Analysis
- **Simulation:** CST Studio Suite, Proteus
- **Wireless Systems:** Microwave Engineering, RF Circuits, Wireless Communication, Antenna Arrays, Beamforming

INDUSTRIAL TRAINING & PRACTICAL EXPOSURE

Biman Bangladesh Airlines – Engineering Division **2024 | 1 Month**

- Collaborated with aircraft maintenance and systems engineers; gained insight into airline engineering operations, aircraft systems, and maintenance workflows.

Bangladesh Air Force (BAF) – Avionics Units **2024 | 1 month**

Units: 205, 207, 214, 216

- Gained hands-on exposure to avionics systems, airborne communication and navigation equipment, and aerospace maintenance procedures; observed radar, electronic warfare, and ground support systems across multiple technical units.

Civil Aviation Authority of Bangladesh (CAAB) **2026**

- Gained exposure to ATC systems, instrument landing systems (ILS), VOR/DME navigation aids, and aviation regulatory infrastructure through facility visits.

Bangladesh Satellite Company Limited (BSCL) **2026**

- Observed satellite ground station communication, telemetry, tracking, and command (TT&C) systems supporting the Bangabandhu-1 satellite.

PROJECTS

AM Superheterodyne Receiver Design for Rangpur Radio Station **2024**

Communication Systems Course Project

- Engineered a complete AM superheterodyne receiver architecture, integrating RF amplifier, local oscillator, mixer, 455 kHz IF stage, bandpass filter, envelope detector, and audio amplifier subsystems; simulated and validated in Proteus.
- Evaluated receiver performance through sensitivity, selectivity, signal quality, and interference rejection analysis — applying RF/communication principles directly relevant to airborne communication and navigation equipment.

Signal-Based Security Monitoring System **2024**

Signals and Systems Course Project

- Designed and implemented a signal-based security monitoring system applying core signal processing concepts for signal acquisition, filtering, and analysis.
- Developed MATLAB-based algorithms for signal acquisition, processing, and real-time monitoring.

REFERENCES

Md. Abu Bakar Siddik
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